

AI/ML Online Test — Task Brief

Task Options (choose one):

1. **Fake News Detection** — Classification task
2. **Movie Recommendation System** — Regression task

Guidelines:

- The candidate is free to choose any method/model of their own for the selected task.
- You are only required to complete **one** task and its corresponding dataset: not both. Choose either (1) Fake News Detection using the Fake-news dataset, or (2) Movie Recommendation System using the MovieLens-100K dataset.
- They must follow 70:10:20 as train:val:test split.
- The candidate must have a strong fundamental understanding of the chosen method, its working, assumptions, and limitations should be explainable, not just implemented.
- Code and reported metrics must be fully reproducible (i.e., consistent results on re-run, with fixed seeds, documented environment/dependencies, and clear data preprocessing steps).
- A final reproducibility checklist must be filled out and submitted to the Dropbox link provided.

Task Description:

Option 1: Fake News Detection (Classification)

Given a news article (title and/or body text), build a model to classify it as **real** or **fake**. This involves text preprocessing (cleaning, tokenization, vectorization/embedding), selecting and training a classification model, and evaluating performance using appropriate metrics (Accuracy, Precision, Recall, F1-score, AUC-ROC).

Option 2: Movie Recommendation System (Regression)

Given user–movie interaction data (ratings), build a model to **predict the rating a user would give to a movie** they haven't rated yet. This involves data preprocessing (handling sparsity, train/val/test splitting at the interaction level), selecting a regression-based recommendation approach, and evaluating using RMSE/MAE. The candidate should justify the modeling approach, and discuss the limitations of the chosen method.

Deliverables:

- Source code (well-documented and runnable end-to-end)
- Chosen model/method with justification
- Evaluation metrics (task-appropriate — e.g., F1/Accuracy/AUC for classification, RMSE/MAE for regression)
- Completed reproducibility checklist

Dataset link: 1) Movielens-100K [\[link\]](#)

2) Fake-news detection [\[link\]](#)

Submission file naming convention:

Place all the code-related files and the readme file (for proper steps on how to reproduce the code provided) inside a folder and compress it (.zip format).

Zip file should be named as **{first_name}_{last_name}.zip**

Example: **Subham_Raj.zip** containing all code-related files along with a readme file.

Note: Failure to follow the instructions given (including naming convention, folder structure, and README requirements) will lead to **disqualification of the submission from evaluation**.

Submission link: <https://www.dropbox.com/request/sfawbsidkm1ey0wtzcjg>